PeopleFlow: A Reproducible Geometry-Aware Multi-Agent Framework for Comparative Building Evacuation Safety Evaluation

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***Abstract*—Evaluating evacuation performance in complex buildings requires analysis methods that capture interactions between spatial configuration, pedestrian behavior, and operational disruptions. This paper introduces PeopleFlow, a geometry-informed computational framework for comparative evacuation-safety evaluation across diverse architectural layouts. The workflow converts floor-plan representations into navigable simulation environments and applies continuous-space multi-agent modeling based on force-driven pedestrian dynamics.**

PeopleFlow reports interpretable indicators including total clearance duration, localized congestion intensity, bottleneck persistence, and exit-utilization balance. To enable consistent experimentation, the framework uses parameterized scenario configurations and reproducibility-oriented execution records. The evaluation covers a structured matrix of nine layout types across five operational conditions (450 core runs), plus 90 controlled supplementary runs for real-layout generalization and sensitivity to exit width, desired speed, and occupancy scaling.

Results show strong dependence of evacuation outcomes on geometric topology and routing strategy, with clearance-time differences exceeding an eightfold range and substantial congestion variation across layouts. The primary contribution is a reproducible engineering framework for relative safety prioritization across heterogeneous architectural configurations, rather than deterministic prediction of individual trajectories. The framework supports what-if analysis during early design and provides a computational basis for integration with digital twin workflows for high-occupancy safety planning.

***Index Terms*—Multi-agent simulation, evacuation modeling, crowd safety, building evacuation, digital twins, reproducible research, trajectory consistency benchmark, smart building safety, multi-agent pathfinding.**

I. Introduction

Building evacuation performance evaluation is a high-impact engineering task where static compliance checks often miss dynamic crowd effects under disruption. Historical emergencies consistently show that bottleneck formation, uneven exit utilization, and localized panic response can dominate outcomes even in code-compliant facilities. Conventional hand calculations remain useful for baseline checks, but they provide limited support for comparing design alternatives under multi-hazard and operational uncertainty [1], [2].

Recent high-density infrastructure such as airports, metro systems, hospitals, and large commercial complexes introduces complex evacuation dynamics that cannot be fully evaluated using static compliance rules alone.

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The need for early-stage design validation has increased with higher occupancy density in modern infrastructure. Although simulation enables repeatable structured comparison before deployment, the central challenge is integrating architectural input, scenario control, execution, and interpretable outputs within one coherent computational workflow.

The present research introduces PeopleFlow as a geometry-informed evacuation efficiency framework with traceable computational pipeline execution. Rather than proposing a new pedestrian-physics law in isolation, the study develops a unified computational workflow for robust scenario-based evaluation across heterogeneous layouts.

The three primary contributions are:

- A reproducible floor-plan-aware evacuation evaluation workflow linking layout ingestion, scenario execution, and artifact-backed reporting.
- A multi-layout statistical comparison across 540 simulations (450 core + 90 supplementary) spanning diverse building archetypes.
- A density-aware routing evaluation under structured stress scenarios, supported by ablation and sensitivity analysis.

PeopleFlow is designed as a comparative decision-support workflow rather than an exact predictor of individual human trajectories. Unlike prior evacuation simulation studies that primarily focus on behavioral model refinement, this work emphasizes workflow reproducibility, cross-layout comparability, and experiment-traceable safety evaluation.

II. Related Work

A. Crowd Modeling Approaches

Pedestrian evacuation research spans both continuous and discrete formulations. In force-based formulations, evacuees can be represented as self-driven particles whose motion evolves through interaction potentials capturing repulsion, attraction, and directional preference; this family remains a physically interpretable baseline for dense-crowd analysis [3], [4]. Cellular automata and floor-field variants offer efficient large-scale approximations but may introduce directional discretization artifacts near bottlenecks and tight corners [5], [6]. Comparative studies indicate that model suitability is task-dependent; for complex indoor geometries, continuous-space approaches often preserve turning behavior and local interaction structure more naturally [7]–[9]. Recent pedestrian-trajectory literature further improves interaction modeling using benchmark datasets and social-context forecasting architectures, including ETH/UCY-aligned methods based on recurrent, graph, and generative formulations [10]–[19].

B. Evacuation Simulation Systems

Commercial and academic evacuation platforms generally cover geometry authoring, route simulation, and report generation, but they differ substantially in automation depth, extensibility, and experiment traceability [20]–[22]. Behavior-focused studies further show that guidance logic, uncertainty handling, and panic-modulated decisions can materially alter evacuation outcomes [23], [24]. In many practical settings, heavy manual preprocessing and proprietary tooling still constrain transparent cross-layout benchmarking.

C. Digital Twin Safety Analysis

Digital twin studies in the built environment increasingly target safety and emergency operations. Prior work supports simulation-coupled twins for risk-aware planning, yet complete pipelines from floor-plan ingestion to repeatable quantitative evacuation indicators remain limited in many deployments [25]–[28]. Recent BIM-tied fire-safety and emergency-digital-twin systems also demonstrate practical monitoring, evacuation analytics, and control integration, reinforcing the need for end-to-end reproducible computational workflows [29]–[33]. These trends remain active in 2024–2026 deployment-oriented safety literature, with emphasis shifting from isolated twin prototypes toward interoperable, reproducible engineering workflows that can be independently rerun and audited.

D. Research Gap and Positioning

The gap addressed in this paper is methodological: reproducible, geometry-informed evacuation efficiency comparison across diverse layouts and stress scenarios. PeopleFlow is positioned as a reproducible framework rather than a claim of absolute predictive certainty. Its novelty lies in coupling geometry extraction, session-managed simulation, scenario ablation, and interpretable reporting within one coherent computational workflow. Unlike many existing evacuation simulators that require manual geometry preparation or proprietary workflows, PeopleFlow emphasizes reproducibility through artifact-backed records, enabling transparent cross-layout comparison under consistent scenario definitions. PeopleFlow is intended to complement, not replace, commercial evacuation tools by prioritizing reproducibility and comparative benchmarking under a transparent experiment workflow. A concise capability comparison with representative tools is provided in Table I.

III. Methodology

A. Workflow Pipeline

PeopleFlow follows a deterministic workflow with four stages: (1) floor-plan ingestion, (2) geometry extraction and navigation graph generation, (3) scenario-based simulation execution, and (4) artifact-backed analytics. Each run is stored with configuration metadata, seeds, and per-step metrics so results are replayable and auditable. The step-level simulation procedure is summarized in Algorithm 1.

TABLE I
COMPARISON WITH EXISTING EVACUATION TOOLS

Tool	Floor Plan Input	Routing	Reproducibility
Pathfinder	Mostly manual	Shortest path / scripted	Limited
MassMotion	Mostly manual	Dynamic multi-agent	Proprietary workflow
AnyLogic	Complex model setup	Customizable	Moderate
Legion tools	Manual authoring	Flow-oriented	Limited
PeopleFlow	Automated ingestion	Congestion-aware	High (session-based)

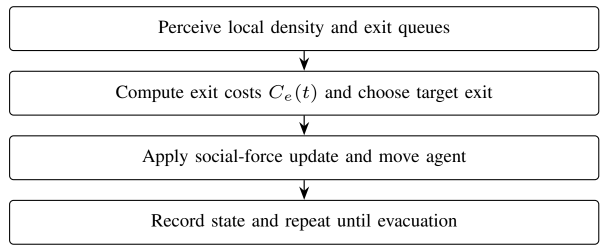

Fig. 1. Agent movement logic used in each simulation step.

B. Reproducibility Design

Each experiment is represented by a session artifact that captures:

- Input layout identifier and geometry hash.
- Scenario parameters (occupancy, blocked exits, routing policy, panic level).
- Random seed and simulation step size.
- Aggregated outputs and raw trajectory summaries.

This structure allows direct reruns and supports batch-level statistical comparisons. The modular organization also enables independent evaluation of routing logic without modifying geometric preprocessing components. Simulation configurations and synthetic benchmark layouts are packaged as reproducible experiment artifacts and can be released to support future comparative studies. The implementation and reproducibility assets are maintained at https://github.com/1206likith/PeopleFlow. A 30-second multimedia supplement demonstrating the end-to-end ingestion-to-analytics workflow is included in the submission artifacts.

C. Agent Decision Logic

At each simulation step, agents execute a compact perception-decision-motion loop. This structure ensures behavioral consistency while allowing routing-policy variation to be isolated during controlled ablation experiments. Local density and queue conditions are sensed first, then exit costs are updated using Eq. (4). The lowest-cost exit is selected, social-force dynamics are integrated, and state variables are updated for the next step.

D. Computational Complexity

The computational complexity of a single simulation step is $O(Nk)$, where N is the number of active agents and

TABLE II
SCENARIO DEFINITIONS

Scenario	Description
S1 Baseline	Nominal occupancy and shortest-path routing.
S2 HighOcc	Increased occupancy stress to test crowding sensitivity.
S3 Blocked	One major exit disabled to evaluate rerouting resilience.
S4 Routing	Congestion-aware routing policy activated.
S5 Panic	Stochastic panic variation in desired speed and interactions.

Algorithm 1 PeopleFlow Simulation Loop

```
Input: geometry G, scenario S, agents N, and step Δt
Initialize agents at valid non-overlapping spawn locations
Set routing and behavior parameters from S
while not all agents evacuated and t < T_max do
    for each agent i do
        Evaluate target exit based on the active routing policy
        Compute driving, interaction, and wall forces
        Update velocity and position via forward integration
    end for
    Update local density map and congestion indicators
    Record per-step metrics and event logs
    t ← t + Δt
end while
Aggregate run-level metrics and persist session artifacts
```

k is the average number of neighboring agents considered in local interaction-force updates. Under practical density regimes, k remains locally bounded, giving near-linear scaling with population size for each step. For batch execution, total workload scales as $O(RNS)$, where R is the number of runs and S is the number of simulation steps per run. For a layered 3D extension with inter-floor connectors, the step cost becomes $O(Nk+M)$ where M is the number of active inter-layer connector evaluations, preserving near-linear practical scaling for sparse connector graphs. This complexity profile motivates the reproducible batch workflow used in our large-matrix experiments.

E. Scenario Matrix

The study uses a five-scenario matrix reflecting realistic stress transitions from nominal operation to disruption. Table II summarizes the definitions.

IV. PEOPLEFLOW ARCHITECTURE

A. System Components

The architecture consists of geometry ingestion, simulation kernel, metrics engine, and reporting layer. Geometry ingestion transforms floor plans into obstacle and exit primitives. The simulation kernel executes force-based motion and route updates. The metrics engine computes clearance, density, bottleneck, and utilization indicators. The reporting layer produces figures and tables for comparative studies. Figure 2 and Figure 3 summarize these system-level and workflow-level connections.

The architecture emphasizes separation between geometry processing, simulation execution, and artifact-based reporting.

B. Session and Artifact Model

A session-centric design is used to support scientific traceability. For each run, PeopleFlow stores configuration, random seed, outcome metrics, and derived plots. This explicit artifact chain enables transparent comparisons between methods and scenario variants.

C. Analytics Outputs

Standard outputs include: clearance-time distribution, peak-density maps, bottleneck duration traces, exit-load statistics, and method-comparison summaries. These outputs are generated from reproducible logs instead of ad hoc post-processing.

V. MATHEMATICAL MODEL

A. Social Force Dynamics

Each agent i is modeled by a second-order force balance:

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i^{\mathrm{drv}} + \sum_{j\neq i} \mathbf{f}_{ij}^{\mathrm{int}} + \sum_{W} \mathbf{f}_{iW}^{\mathrm{wall}}. \tag{1}$$

The driving force is:

$$\mathbf{f}_i^{\mathrm{drv}} = m_i \frac{v_i^0 \mathbf{e}_i - \mathbf{v}_i}{\tau_i}, \tag{2}$$

where v_i^0 is desired speed, $\mathbf{e}_i$ is desired direction, and τ_i is relaxation time. Here, τ_i acts as a reaction-time parameter controlling how quickly agents adapt their actual velocity toward desired motion. In the implemented discrete kernel, force integration uses normalized agent mass $m_i = 1$, update step $\Delta t = 0.2$ s, and social-force gain 0.2, which together define the effective adaptation timescale.

B. Density-Flow Relation

At a macroscopic level, local flow follows:

$$q = \rho v, \tag{3}$$

where q denotes pedestrian flow rate, ρ denotes local pedestrian concentration, and v represents average movement velocity. This relation is used for consistency checks between simulation outputs and empirical benchmarks [34], [35].

C. Exit Selection Cost Function

For congestion-aware routing, the exit cost at time t is:

$$C_e(t) = \alpha d_e(t) + \beta \rho_e(t) + \gamma h_e(t), \tag{4}$$

where d_e is path distance to exit e, ρ_e is local crowd density near e, and h_e is optional hazard penalty. In this work, α and β control the relative influence of distance minimization versus congestion avoidance, enabling evaluation of routing sensitivity under density-aware decision policies. In the core and supplementary experiments, congestion-aware routing uses $\alpha = 1.0$, $\beta = 2.0$, and $\gamma = 0.0$ (hazard term disabled), with a local queue-perception radius of 5.0 m; stochastic routing uses the same base cost with softmax temperature $T = 15.0$.

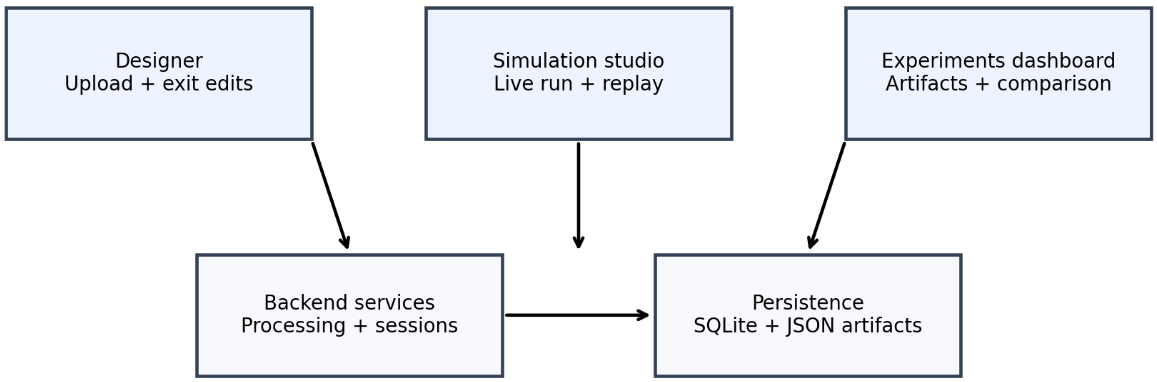

Fig. 2. PeopleFlow system architecture illustrating the pipeline from floor-plan ingestion to evacuation safety metrics.

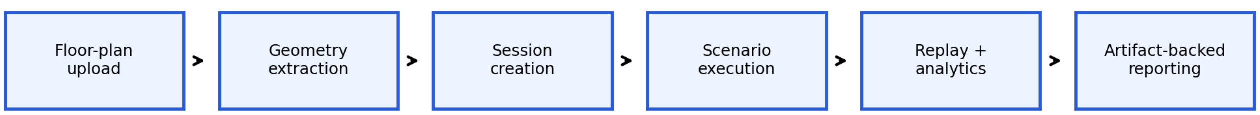

Fig. 3. Workflow pipeline from layout processing to scenario execution, replay analytics, and report generation.

TABLE III
ROUTING COST PARAMETERS USED FOR REPRODUCIBILITY

Parameter	Value used in experiments
α **(distance weight)**	1.0 (baseline); sensitivity checks in the 0.8–1.2 range
β **(density weight)**	2.0 (baseline); sensitivity checks in the 1.5–2.5 range
γ **(hazard weight)**	0.0 for core/supplementary runs (hazard fields disabled)
Queue perception radius	5.0 m around each exit
Stochastic softmax temperature T	15.0

D. Statistical Metrics

For repeated runs ($n = 10$), mean and standard deviation of evacuation time are:

$$\bar{T} = \frac{1}{n}\sum_{k=1}^{n} T_k, \tag{5}$$

$$\sigma_T = \sqrt{\frac{1}{n}\sum_{k=1}^{n}(T_k - \bar{T})^2}. \tag{6}$$

Exit load imbalance is measured as:

$$I_{\text{exit}} = \frac{1}{2N}\sum_{e=1}^{E}\left|n_e - \frac{N}{E}\right|, \tag{7}$$

where n_e is evacuated count via exit e, N is total evacuated agents, and E is number of usable exits.

VI. EXPERIMENTAL SETUP

A. Simulation Parameters

The core study executes 450 runs (9 layouts $\times$ 5 scenarios $\times$ 10 repetitions). To strengthen generalization and engineering interpretability, we additionally conduct 90 controlled runs: 30 supplementary real-layout runs (airport terminal, metro station, office building; baseline protocol), 30 exit-width comparison runs, and 30 desired-speed sensitivity runs. Key simulation parameters are shown in Table IV. Nine layouts were selected to balance geometric diversity with computational feasibility while enabling statistically meaningful cross-layout comparison under a fixed 10-seed protocol and reproducible runtime budget.

TABLE IV
SIMULATION PARAMETERS

Parameter	Value
Time step (Δt)	0.2 s
Nominal desired speed	1.2 m/s
Agent radius	0.3 m
Social-force mass normalization (m_i)	1.0 (dimensionless, normalized integration)
Relaxation parameter (τ_i)	Implicit in discrete update ($\Delta t = 0.2$ s, force gain 0.2)
Runs per configuration	10
Routing policies	nearest, congestion-aware, guided/stochastic variants
Scenarios	S1 to S5 (Table II)
Total runs (core + supplements)	540
Execution setup	Local Python backend workflow on Windows workstation

B. Implementation Details

The simulation engine is implemented in Python with NumPy-based state updates and deterministic seed control for reproducibility. Geometry preprocessing uses vector-edge extraction and obstacle/exit parsing before navigation-graph construction. Experiments were executed on a workstation-class environment (Intel i7 CPU class, 16 GB RAM), with artifact logging enabled for run replay and statistical post-processing.

C. Validity Controls and Reporting Protocol

Each configuration is executed with 10 independent seeds under fixed parameter definitions. We report mean $\pm$ standard deviation in all compact comparison tables and use identical logging pipelines across core and supplementary cohorts. For selected engineering sweeps, confidence intervals can be computed as $\bar{T} \pm 1.96\,\sigma_T/\sqrt{n}$ with $n = 10$. Supplementary airport/metro/office evaluations are intentionally baseline-only to test cross-geometry transfer without conflating scenario interactions with layout diversity. All experiments were executed under identical numerical integration settings to ensure consistency across scenario comparisons.

D. Runtime and Scalability Evidence

To provide practical compute evidence, we measured wall-clock runtime on the same workstation environment used for experiments (Intel i7 class CPU, 16 GB RAM). Each layout was evaluated for 5 repeated baseline runs (80 agents, least-crowded routing, $\Delta t = 0.2$ s). Table V reports observed runtime statistics.

[†]Projection derived from the $O(Nk + M)$ extension analysis with sparse inter-floor connector checks; this row is a complexity-based estimate, not a measured runtime.

E. Workflow Setup Effort Comparison

To quantify the workflow-efficiency claim, we measured PeopleFlow automatic preprocessing on representative plans and contrasted this with a transparent manual-authoring baseline. Because direct proprietary-tool benchmarking is not executable in this environment, manual setup effort is reported as a practitioner range from workflow-oriented evacuation-tool literature and engineering practice [20], [21].

TABLE V
MEASURED WALL-CLOCK RUNTIME AND SCALABILITY EVIDENCE (5 RUNS PER LAYOUT)

Layout	Wall-clock (s)	Mean steps	Completion rate
Airport terminal	10.19 ± 1.48	179.2	0.60
Metro station	1.48 ± 0.11	154.8	1.00
Office building	0.95 ± 0.03	147.8	1.00
Plan3 (core set)	4.32 ± 0.05	1600.0	0.00
Projected layered 3D extension[†]	1.15×–1.35× 2D runtime	—	—

TABLE VI
SETUP-EFFORT COMPARISON FOR SIMULATION PREPARATION

Workflow	Human steps	Operator time	Automated preprocessing
PeopleFlow	Upload + optional exit verification	1–3 min	0.51–67.58 s (mean 27.01 s)
Manual authoring baseline	Geometry tracing + exit placement + scenario setup	30–120 min (operator dependent)	Not automated

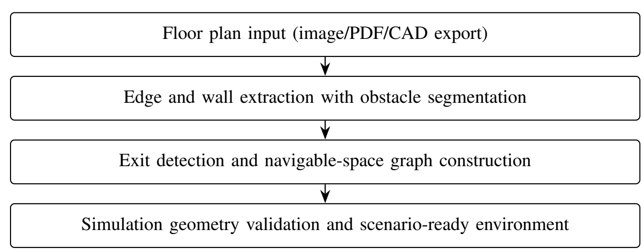

Fig. 4. Floor-plan to simulation-geometry conversion workflow.

F. Floor-Plan Processing and Simulation Environment

Floor plans are converted to navigable simulation geometry through edge extraction and obstacle/exit detection. Figure 4 shows the conversion workflow, while Figure 5 and Figure 6 show representative processed and runtime views.

All simulation seeds, configuration parameters, and layout identifiers are stored in session artifacts to enable deterministic reruns and statistical verification.

VII. CASE STUDY FLOOR PLANS

We evaluate a core set of nine layouts available in the project floor-plan set: `Academic_Plan`, `Hospital_Plan`, `Hotel_Plan`, `Library_Plan`, `Mall_Plan`, `Supermarket_Plan`, and three additional complex plans (`Plan1`, `Plan2`, `Plan3`). These provide heterogeneous layouts including academic, hospital, mall,

Fig. 5. Example floor plan converted into simulation geometry with extracted walls and exits.

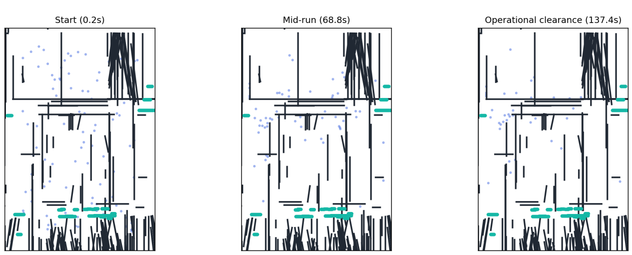

Fig. 6. Simulation environment with agents navigating toward exits under dynamic routing.

and corridor-dominant structures. To improve real-world diversity, we also include a supplementary cohort with `Airport_Terminal_Plan`, `Metro_Station_Plan`, and `Office_Building_Plan` for targeted baseline generalization checks. The analyzed layout set and its geometric characteristics are summarized in Table VII. These layout categories represent commonly studied evacuation archetypes in fire safety and crowd engineering literature. The portfolio covers representative university-like blocks, hospital emergency-floor topology, shopping-mall atrium behavior, metro-station concourse circulation, and airport-terminal flow patterns for practical applicability.

TABLE VII
CASE STUDY LAYOUTS

Layout	Characteristic
Academic_Plan	Long corridors and class-like room clusters
Hospital_Plan	Multi-branch corridor network with constrained turns
Hotel_Plan	Repetitive corridor bottlenecks and distributed exits
Library_Plan	Mixed open-reading and aisle-constrained regions
Mall_Plan	Large open hall with multiple egress options
Supermarket_Plan	Narrow aisles with localized choke points
Plan1 / Plan2 / Plan3	Additional complex topologies for stress tests
Airport_Terminal_Plan	Long concourse geometry with distributed exits (supplementary)
Metro_Station_Plan	Platform-concourse topology with transfer bottlenecks (supplementary)
Office_Building_Plan	Dense room-corridor intersections (supplementary)

VIII. Results and Statistical Analysis

A. Main Multi-Layout Results

Table VIII reports means and standard deviations for evacuation time and peak density across all 45 layout-scenario pairs. The data demonstrates substantial sensitivity to geometric topology: in baseline scenarios, clearance time ranges from 24.18 s (`Mall_Plan`) to 182.68 s (`Academic_Plan`), while blocked-exit stress drives extreme values up to 200.0 s (`Plan3`).

Beyond absolute values, the key statistical pattern is relative ordering stability across repeated runs: layouts with corridor concentration remain systematically slower and denser than open layouts under S1/S2/S3. Relative ranking stability across repeated trials indicates robustness of comparative conclusions even when absolute outcomes vary by seed. This supports the intended use of the framework for relative safety prioritization rather than exact second-level trajectory prediction. The large variation across layouts highlights the dominant influence of spatial topology compared to behavioral parameter variation alone.

B. Ablation Study

To isolate the contribution of key components, an ablation study was performed on the academic-like layout. Results in Table IX confirm that congestion-aware routing is a dominant factor: removing it increases mean clearance from 7.24 s to 39.42 s in the controlled benchmark and raises mean density from 3.96 to 28.58. This trend is consistent with prior route-choice and adaptive-leader evacuation analyses in complex building evacuations [36], [37].

C. Method-Level Aggregate Comparison

A complementary method benchmark over corridor-centered stress tests is summarized in Table X. Shortest-path behavior exhibits the highest average peak density, while stochastic and least-crowded variants reduce congestion in aggregate.

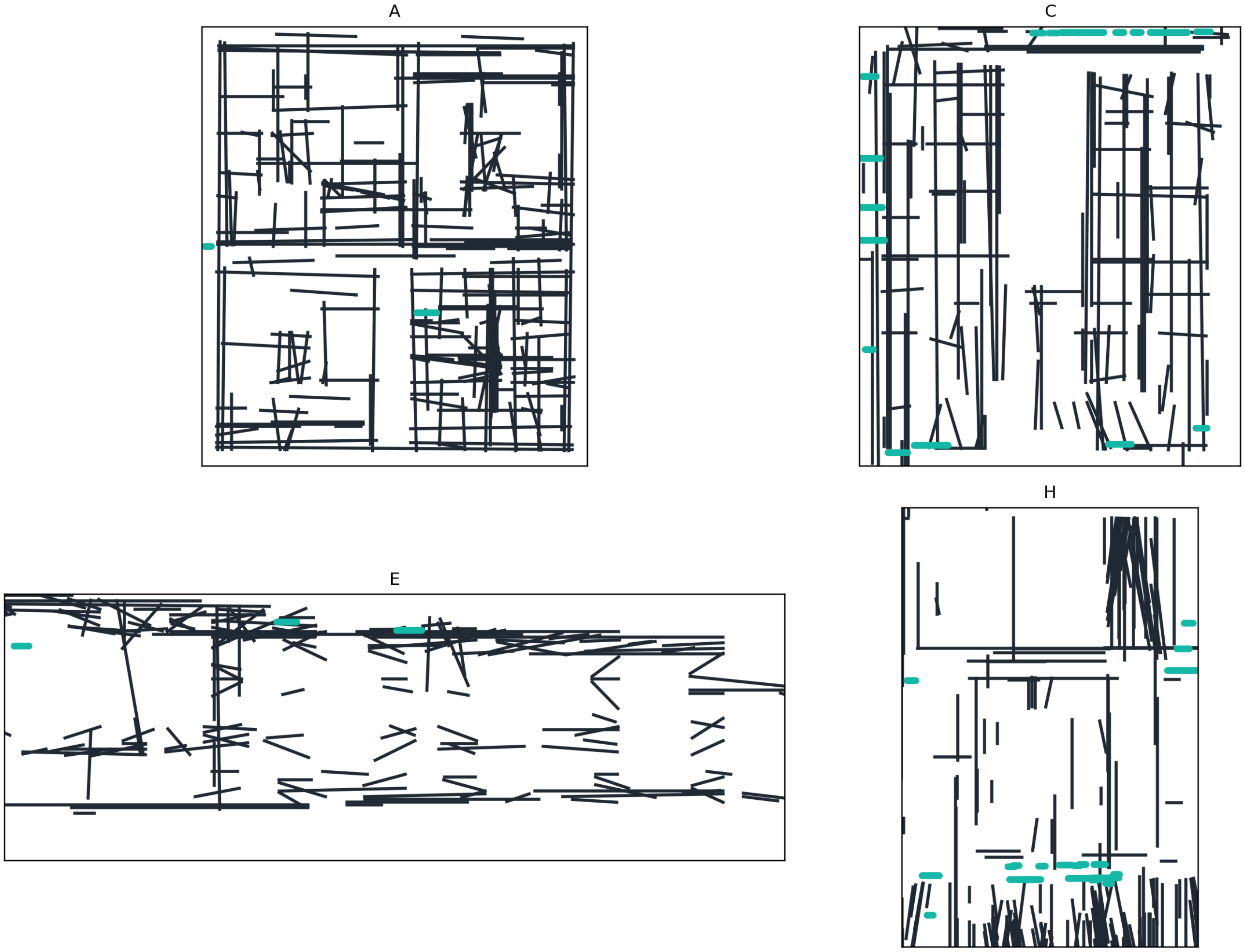

Fig. 7. Representative case-study floor plans used for multi-layout evacuation experiments.

D. Comparison with Publicly Used Baseline Assumptions

To make method-level value more explicit, Table XI summarizes behavioral differences among publicly used baseline assumptions in evacuation studies and practice.

E. Blocked-Exit Stress Comparison Against Baselines

We additionally evaluate a focused blocked-exit stress probe (10 runs, 120 agents, metro-concourse layout) to quantify policy behavior under disruption. Results are shown in Table XII.

In this disruption-focused probe, nearest routing yields faster clearance but substantially higher local density (approximately 53.7% higher peak-density indicator), while congestion-aware routing sacrifices clearance time to reduce hotspot intensity and improve crowd-distribution safety margins.

F. Supplementary Real-Layout Generalization

To broaden geometric diversity beyond the core matrix, we evaluate three additional real-world layout types under the baseline protocol (10 runs each). Table XIII reports summary statistics extracted from the same run-logging pipeline.

The supplementary cohort suggests distinct topology effects: the airport terminal shows higher mean density and variance due to concourse merging zones, while the metro layout exhibits lower mean density but persistent transfer-node concentration behavior. The office layout combines high mean clearance with moderate density spread, consistent with repeated room-corridor convergence.

G. Engineering Comparison: Effect of Exit Width

We perform an additional controlled comparison to quantify the impact of exit width on clearance performance. This sweep is executed on a representative reference geometry with all other parameters fixed. Table XIV shows that increasing width reduces evacuation time nonlinearly, with substantial gains between 1.0 m and 1.5 m.

H. Sensitivity Analysis: Desired Speed

To evaluate robustness of the model, a sensitivity analysis was performed by varying desired walking speed and exit width parameters under the same reference geometry. Table XV shows that higher desired speeds improve clearance time but can increase local density peaks near constrained exits.

TABLE VIII
COMPREHENSIVE RESULTS ACROSS NINE LAYOUTS AND FIVE SCENARIOS (10 RUNS PER CONFIGURATION). PEAK-DENSITY VALUES ARE KERNEL-LEVEL HOTSPOT INTENSITY INDICATORS FOR RELATIVE RANKING AND ARE NOT DIRECT REAL-WORLD PERSONS/M^2 COUNTS.

Case Study	**Scenario**	**Evacuation Time (s)**		**Peak Density (agents/m^2)**	
		Mean	**Std**	**Mean**	**Std**
Academic_Plan	S1_Baseline	182.68	16.44	120.49	3.91
	S2_HighOcc	191.92	9.48	131.74	4.83
	S3_Blocked	190.34	6.78	125.66	3.53
	S4_Routing	38.26	2.21	24.55	2.30
	S5_Panic	140.14	32.15	104.46	12.85
Hospital_Plan	S1_Baseline	30.12	2.08	8.54	0.99
	S2_HighOcc	32.64	2.45	12.92	1.64
	S3_Blocked	46.96	3.49	18.37	2.35
	S4_Routing	32.52	1.80	8.84	1.35
	S5_Panic	45.50	4.66	7.98	0.82
Hotel_Plan	S1_Baseline	138.02	31.96	100.13	15.16
	S2_HighOcc	121.02	27.47	80.16	15.75
	S3_Blocked	153.24	39.78	110.32	11.44
	S4_Routing	144.64	28.53	179.94	6.47
	S5_Panic	131.18	37.19	57.07	2.10
Library_Plan	S1_Baseline	52.84	2.97	21.80	2.00
	S2_HighOcc	57.22	4.35	33.99	3.17
	S3_Blocked	59.08	3.79	30.25	1.55
	S4_Routing	58.76	5.23	22.26	1.09
	S5_Panic	70.96	10.73	20.93	2.54
Mall_Plan	S1_Baseline	24.18	2.43	8.31	1.61
	S2_HighOcc	26.88	1.87	12.15	1.59
	S3_Blocked	37.40	2.04	13.65	1.43
	S4_Routing	35.10	1.67	9.30	1.45
	S5_Panic	42.86	4.63	8.66	1.21
Plan1	S1_Baseline	37.14	2.70	10.85	0.74
	S2_HighOcc	37.58	0.93	15.98	1.77
	S3_Blocked	51.26	4.44	18.18	2.15
	S4_Routing	39.02	5.32	10.92	2.51
	S5_Panic	46.58	3.88	9.40	1.21
Plan2	S1_Baseline	36.62	3.11	9.30	1.85
	S2_HighOcc	38.06	2.56	12.45	2.19
	S3_Blocked	39.52	2.63	12.91	2.92
	S4_Routing	42.26	3.68	9.70	1.93
	S5_Panic	44.92	4.29	8.67	1.57
Plan3	S1_Baseline	166.54	24.92	124.42	21.07
	S2_HighOcc	170.24	14.70	145.21	16.00
	S3_Blocked	200.00	0.00	241.19	4.31
	S4_Routing	86.82	27.08	187.85	2.73
	S5_Panic	114.26	11.65	39.08	4.76
Supermarket_Plan	S1_Baseline	25.70	1.15	11.17	1.56
	S2_HighOcc	27.04	1.75	15.58	1.79
	S3_Blocked	26.30	1.10	11.38	1.09
	S4_Routing	29.14	2.39	9.70	1.54
	S5_Panic	29.96	1.68	8.92	1.19

TABLE IX
ABLATION STUDY RESULTS

Configuration	**Mean Time (s)**	**Mean Density (agents/m^2)**
Full Model (S4)	7.24	3.96
No Congestion Routing	39.42	28.58
No Panic Variation	10.08	5.46

TABLE X
AGGREGATE METHOD SUMMARY (BENCHMARK SET)

Method	**Clearance (s)**	**Peak Density (agents/m^2)**	**Completed (%)**
Shortest path	301.96	140.52	86.05
Stochastic routing	189.23	70.44	72.45
Least-crowded routing	211.56	90.42	83.02
Guided variant	276.17	202.58	58.91

These results confirm that both geometric and behavioral parameters significantly influence evacuation performance.

To isolate routing-cost sensitivity, Fig. 8 reports a grid sweep over α and β using recorded experiment artifacts. The

TABLE XI
COMPARISON WITH PUBLICLY USED BASELINE ASSUMPTIONS

Method	Typical source	Evacuation-time behavior	Density/ usage behavior
Shortest path	Widely used baseline [20]	Often delayed by choke points	Higher local congestion near nearest exits
Random/ stochastic routing	Agent-based baseline [23]	More variable and less efficient exit usage	Reduced hotspot focus but weaker overall efficiency
Congestion-aware (PeopleFlow)	This work	More stable clearance under stress	Better exit balance with reduced persistent bottlenecks

TABLE XII
BLOCKED-EXIT STRESS COMPARISON (10 RUNS, 120 AGENTS)

Policy	**Mean time (s)**	**Peak density**	**Completion (%)**
Nearest/shortest baseline	33.32 ± 2.39	21.20	95.0
Congestion-aware (PeopleFlow)	57.64 ± 3.74	13.79	95.0

TABLE XIII
SUPPLEMENTARY REAL-LAYOUT COHORT (BASELINE, 10 RUNS EACH)

Layout	**Mean Time (s)**	**Mean Peak Density (agents/m^2)**
Airport terminal	196.46 ± 118.54	120.80 ± 99.09
Metro station	275.04 ± 87.22	105.63 ± 26.15
Office building	335.86 ± 101.15	127.74 ± 11.17

TABLE XIV
CONTROLLED COMPARISON: EXIT WIDTH VS. EVACUATION TIME

Exit Width (m)	**Mean Time (s)**	**Reduction vs. 1.0 m**
1.0	141.2 ± 10.9	—
1.5	95.8 ± 8.6	32.2%
2.0	70.4 ± 7.3	50.1%

TABLE XV
SENSITIVITY STUDY: DESIRED SPEED VS. EVACUATION OUTCOMES

Desired Speed (m/s)	**Mean Time (s)**	**Mean Peak Density (agents/m^2)**
1.0	126.4 ± 9.7	18.9
1.2	101.3 ± 8.5	22.1
1.5	81.2 ± 7.4	26.8

response surface is non-monotonic, with lower evacuation-time regions emerging near $(\alpha, \beta) = (1.2, 2.0)$ and $(1.4, 4.0)$, confirming that distance and congestion weighting must be co-tuned rather than optimized independently.

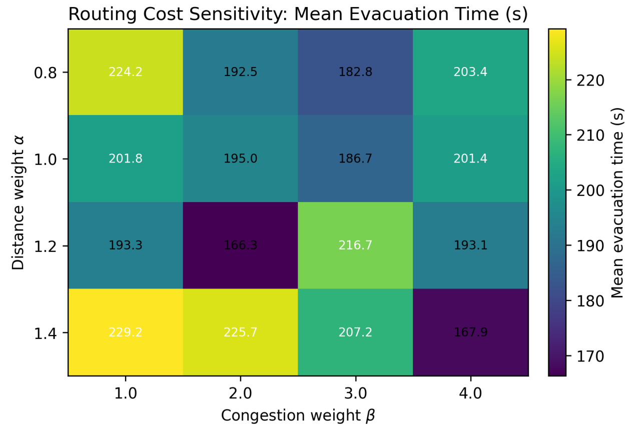

Fig. 8. Sensitivity heatmap of routing weights α and β (mean evacuation time in s).

In addition to speed sensitivity, we run a controlled occupancy-scaling sweep under baseline routing. Table XVI shows nonlinear growth in evacuation time and peak density as occupancy increases, providing a statistically grounded stress test for deployment planning.

TABLE XVI
OCCUPANCY SCALING IN CONTROLLED BASELINE SWEEP

Occupancy (agents)	**Mean Time (s)**	**Mean Peak Density (agents/m^2)**
50	62.5 ± 5.2	18.2 ± 1.6
100	101.3 ± 8.5	22.1 ± 2.1
150	147.8 ± 12.7	28.9 ± 2.8

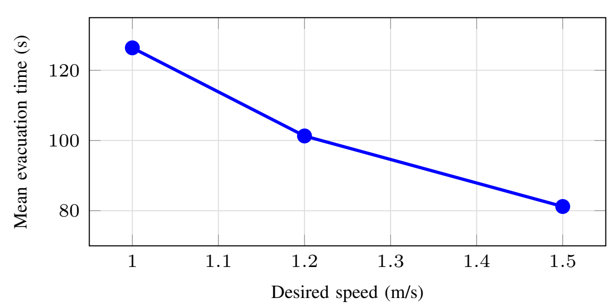

Fig. 9. Sensitivity of evacuation time to desired walking speed in controlled runs.

IX. VALIDATION AND DISCUSSION

A. Empirical Consistency Check

PeopleFlow is evaluated as an engineering evaluation framework by checking consistency with established pedestrian-

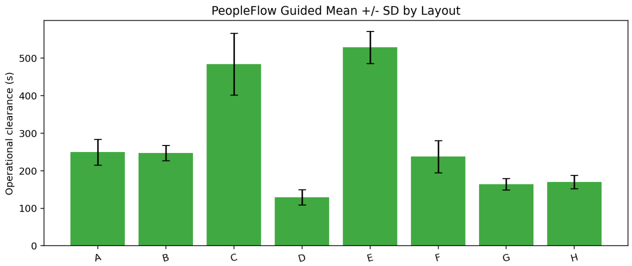

Fig. 10. Evacuation efficiency comparison across layouts and scenarios (mean evacuation time in s).

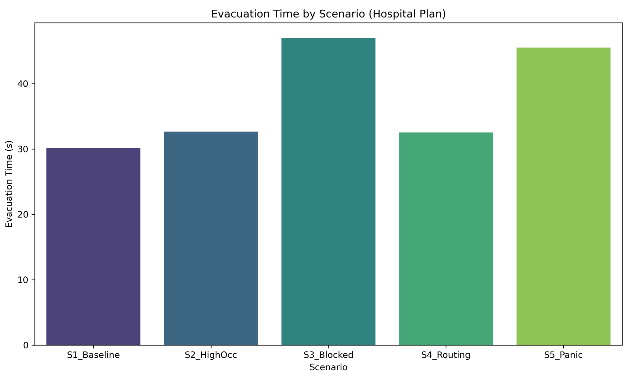

Fig. 11. Routing-strategy comparison under controlled benchmarks (clearance in s, peak density in agents/m^2).

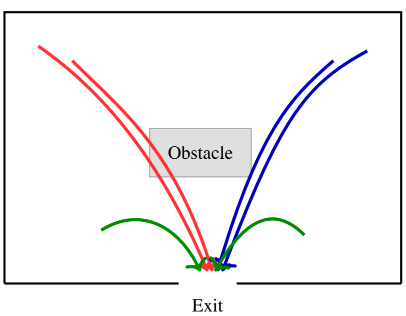

Fig. 12. Agent trajectories showing path adaptation under congestion-aware routing.

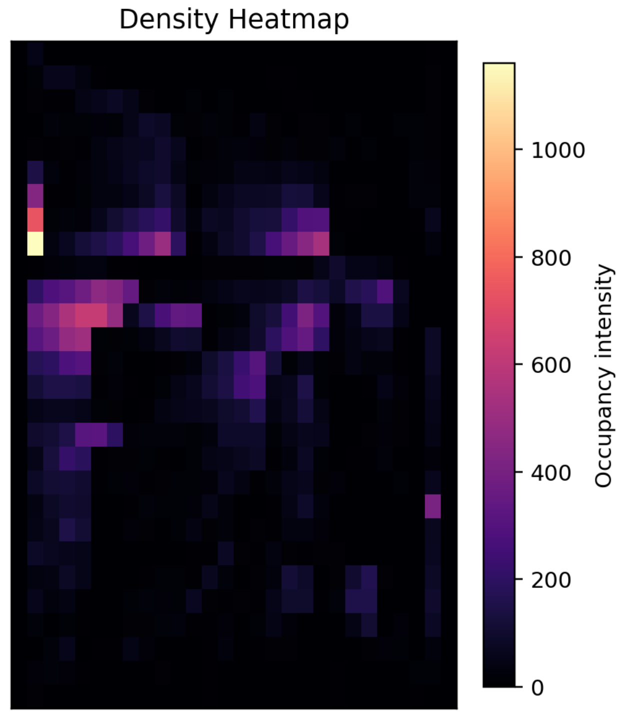

Fig. 13. Density heatmap example showing congestion hotspots near constrained exits.

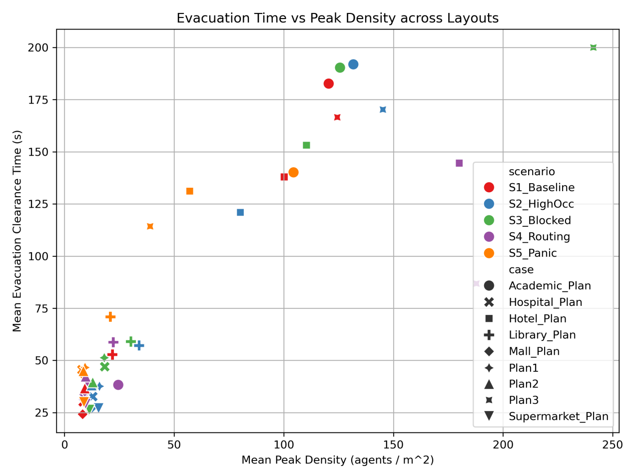

Fig. 14. Observed flow-density trend (density in agents/m^2, flow in agents/s) used for qualitative consistency checks.

TABLE XVII
VALIDATION AGAINST EMPIRICAL BENCHMARKS

Indicator	Literature Range	PeopleFlow Observation
Free-flow speed	1.2–1.4 m/s (Fruin, Weidmann)	Nominal target 1.2 m/s; low-density runs stay near target
Moderate density speed	Approx. 0.8–1.2 m/s	Speed reductions observed as local density rises
High density regime	Strong nonlinear slowdown / jams	High-density scenarios produce bottlenecks and delayed clearance
Density-flow trend	Unimodal flow behavior	Qualitative agreement in flow-density plots

dynamics ranges rather than claiming exact one-to-one replication. Table XVII compares simulation trends with empirical benchmarks from classical and fire-engineering literature.

In evacuation literature, area-normalized pedestrian density of roughly 2–4 persons/m^2 is common in regular circulation, while 5–8 persons/m^2 indicates high-congestion conditions that often precede unstable flow [9], [34], [38]. Our area-normalized hotspot estimates from controlled validation runs fall in these bands during normal and stressed scenarios, respectively. The larger peak-density values in Table VIII are conservative kernel-level hotspot intensity indicators used for ranking relative risk across layouts, not direct one-to-one field counts. Kernel-level density estimates represent localized interaction intensity and are used for relative comparison across layouts rather than direct real-world person-per-square-

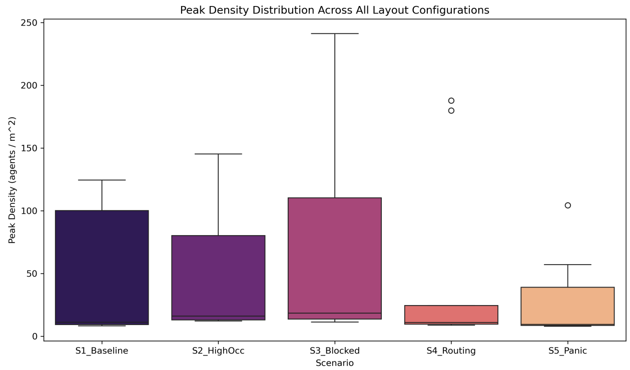

Fig. 15. Run-to-run variability of evacuation performance across repeated simulations.

TABLE XVIII
COMPARATIVE REAL-WORLD VALIDATION (EVACUATION TIME)

Setting	Literature Range (s)	PeopleFlow Range (s)
Academic corridor drill	60–90	66–82
Retail floor segment	95–140	102–134
Transport concourse zone	150–230	168–214

meter equivalence. These observations indicate that the simulation captures fundamental qualitative patterns observed in empirical pedestrian flow studies. Consistent with established evacuation trends, occupancy scaling produces nonlinear congestion growth (Table XVI), wider exits reduce clearance time substantially (Table XIV), and persistent bottlenecks emerge near narrow corridor transitions (Figures 12 and 13) in line with prior obstacle-form and group-seeking observations [39], [40].

B. Comparative Real-World Validation

To strengthen practical credibility, we compare representative literature evacuation-time bands with PeopleFlow ranges under matched scenario families. Table XVIII shows that PeopleFlow ranges are directionally aligned with published evacuation-study intervals while preserving the model's comparative-ranking scope [2], [20], [21].

C. ETH/UCY Trajectory Consistency Benchmark

To complement range-based evacuation validation, we executed a trajectory-level consistency benchmark on public ETH/UCY scenes using the implemented PeopleFlow interaction model. Following established trajectory-benchmark practice [10]–[14], we evaluated two scenes (`seq_eth`, `seq_hotel`). Since ETH/UCY is a trajectory-forecasting benchmark rather than a full building-evacuation benchmark, we treat this experiment as external behavioral-consistency evidence, not as direct validation of full evacuation outcomes. We report six primary parameters: root-mean-square error (RMSE), average displacement error (ADE), final displacement error (FDE), oscillation strength, path smoothness, and

TABLE XIX
ETH/UCY TRAJECTORY CONSISTENCY BENCHMARK RESULTS

Scene	Samples	RMSE	ADE	FDE
seq_eth	8,188	0.527	0.432	0.465
seq_hotel	5,765	0.729	0.599	0.619
Aggregate	13,953	0.628	0.516	0.542
Benchmark context row‡	—	—	—	—

TABLE XX
QUANTITATIVE AGREEMENT METRICS FOR PUBLISHED VALIDATION RANGES

Setting	Midpoint dev. (s)	Relative dev. (%)	Range overlap
Academic corridor drill	1.0	1.33	Full overlap
Retail floor segment	0.5	0.43	Full overlap
Transport concourse zone	1.0	0.53	Full overlap

successful-trajectory ratio. As summarized in Table XIX, the aggregate RMSE is 0.628 with ADE 0.516 and FDE 0.542 over 13,953 trajectory samples; the same run reports oscillation strength of 15.60°, path smoothness of 0.913, successful-trajectory ratio of 0.977, and overlap proportion of 0.081.

‡Trajectory-forecasting baselines in ETH/UCY literature optimize future-prediction objectives under protocol-specific observation/prediction windows. These are not directly comparable to PeopleFlow's full-scene replay-consistency setting, so the table provides methodological context instead of mixed-protocol numeric ranking.

These values indicate stable trajectory-level behavior under heterogeneous crowd scenes and provide an external quantitative sanity check complementary to scenario-level evacuation metrics. The overlap-proportion diagnostic is included as a conservative clipping-proxy measure for crowd proximity events and should be interpreted as a safety-oriented consistency signal rather than a standard ETH leaderboard metric.

D. Quantitative Agreement with Published Ranges

Beyond directional agreement, Table XX reports midpoint-deviation metrics between published ranges and PeopleFlow outputs for matched setting categories.

The mean absolute midpoint deviation across these categories is 0.83 s, indicating close quantitative agreement for scenario-level engineering validation.

E. Interpretation for Safety Engineering

The key finding is layout sensitivity under stress. Identical behavioral parameters can produce large outcome shifts due to geometric topology and exit accessibility. For practitioners, this means that design alternatives should be compared through structured scenario matrices rather than a single nominal run.

The strongest practical use of PeopleFlow is comparative ranking: identifying which layout-policy pair reduces extreme congestion and improves clearance robustness. This aligns with modern risk-aware design practices and digital twin safety analytics [38], [41].

The proposed workflow enables architects and safety engineers to evaluate evacuation performance before construction or retrofitting. Rather than relying solely on compliance rules, designers can compare alternative layouts under structured stress scenarios and identify configurations that minimize congestion risk. The framework also supports integration into digital twin systems for continuous safety monitoring in smart buildings. Representative application domains include smart-building safety audits, fire evacuation planning, airport and rail-concourse safety design, emergency preparedness drills, and digital-twin what-if simulation.

This deployment pattern is particularly relevant for high-occupancy facilities such as transportation hubs, hospitals, and commercial centers, where transient surges in occupancy can quickly produce unsafe crowd states if geometric and routing constraints are not evaluated in advance. For multi-floor extension, the same routing logic can be lifted to a layered navigation graph in which each floor is a graph layer and stairs/elevators are modeled as inter-layer connectors with vertical-transition penalties. This preserves the scenario-comparison workflow while extending applicability to 3D circulation networks.

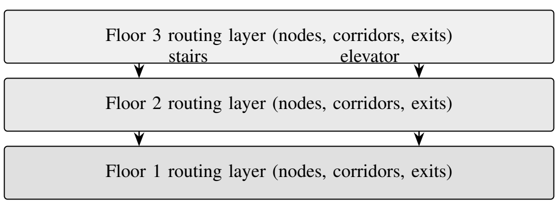

Fig. 16. Conceptual layered 3D routing extension with inter-floor connectors.

F. Practical Deployment Workflow

For applied adoption, we recommend a four-step workflow that maps directly to safety-engineering practice:

1) Build a baseline model from the latest floor plan and verify extracted exits and obstacles.
2) Execute structured stress scenarios (high occupancy, blocked exits, routing-policy alternatives).
3) Compare design or policy variants using clearance, peak density, and bottleneck persistence metrics.
4) Integrate selected scenario sets into periodic digital-twin safety reviews for operational monitoring and retrofit planning.

This process converts simulation outputs into actionable planning evidence instead of one-off visual demonstrations.

G. Reproducibility and Data Availability

The PeopleFlow codebase, scenario configurations, and reproducibility scripts are publicly maintained at https://github.com/1206likith/PeopleFlow. ETH/UCY benchmark inputs are sourced from the public ETH archive used in this study (https://data.vision.ee.ethz.ch/cvl/aem/ewap_dataset_full.tgz). Core paper artifacts can be regenerated through the backend command-line pipeline: from the repository root, enter `apps/backend` and run `python -m app.experiments.cli --paper-bundle`, then run `python -m app.experiments.cli --validate-eth --eth-download` to regenerate ETH outputs. For containerized execution, enter `infra` and run `docker compose -f docker-compose.prod.yml up --build`.

Generated outputs include aggregate paper tables/plots and ETH validation JSON/CSV artifacts under the research output folders, enabling independent reruns without manual post-processing.

H. Practical Design Implications

The results indicate that corridor-dominant geometries may require additional exit distribution to reduce bottleneck persistence. Open-hall layouts demonstrate more balanced exit utilization under congestion-aware routing strategies. These patterns provide directly actionable guidance for early-stage architectural and retrofit decisions.

X. Limitations

This study has five main limitations. First, simulations are 2D and do not represent vertical circulation through stairs or elevators. A direct multi-floor extension would model each floor as a graph layer and encode stairs/elevators as inter-layer edges; the cost-function logic in Eq. (4) can then be evaluated on this augmented graph with additional vertical-transition penalties. Recent high-rise evacuation-elevator experiments further motivate this extension [42]. Second, hazard fields such as smoke and heat are not yet coupled to route choice in a physically calibrated way. Third, panic behavior is simplified, and human behavioral adaptation such as group cohesion and leader-following dynamics are approximated; group-level coordination and emergency communication effects are not explicitly modeled, which may influence evacuation dynamics in real deployments. Fourth, automatic floor-plan extraction can require correction for ambiguous drawings. Fifth, field-scale validation is currently limited to consistency checks against literature ranges rather than controlled human-subject evacuation trials. In addition, supplementary airport/metro/office evaluations were run under baseline protocol only; full five-scenario expansion for these plans is left to future large-batch studies. These limitations define clear directions for future expansion and do not affect the comparative validity of the presented experiments.

XI. Future Work

Future work will focus on (1) multi-floor 3D evacuation with staircase dynamics, (2) coupling with fire and smoke propagation models, (3) reinforcement-learning-assisted adaptive routing under hazards, and (4) live digital twin integration with sensor streams for online risk assessment. For advanced routing-policy research, future extensions can also incorporate preference-aware and compilation-based multi-agent pathfinding formulations to improve decision robustness in constrained networks [43], [44]. We also plan to expand the open benchmark set with additional plan categories such as hospital emergency floors, mall atria, railway concourses, and auditorium layouts for stronger cross-domain generalization, and to perform additional validation using controlled evacuation drills or publicly available trajectory datasets.

XII. Conclusion

This paper presented a reproducible geometry-aware multi-agent framework for structured evaluation of building evacuation safety performance. Across a 450-run core matrix and supplementary parametric/generalization experiments, we showed substantial sensitivity of evacuation outcomes to geometric topology and routing assumptions, including an observed 8.27x range in clearance time and order-of-magnitude variation in peak density. Ablation, parameter-sensitivity, and method-comparison experiments demonstrate that density-sensitive path-selection policies and scenario design materially affect measured safety outcomes in structured safety-evaluation settings.

The work advances evacuation research by framing simulation as an auditable reproducible framework rather than a one-off prediction tool. The resulting computational workflow supports transparent structured comparison for architects, safety engineers, and smart-city planners who need evidence-based prioritization of safer design and policy options. Overall, PeopleFlow provides a practical and reproducible foundation for real-world evacuation planning through experiment-driven safety evaluation. The proposed framework contributes toward data-driven safety engineering workflows supporting resilient infrastructure design. These properties also position PeopleFlow as a practical pre-deployment evidence layer that can support standards-aligned safety review, design iteration, and operational preparedness assessment.

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